

Drugs Used In The Treatment Of Cardiovascular Disorders

Module Outline

- Drugs used in Hypertension
- Drugs used in Angina Pectoris
- Drugs Used in Heart Failure
- Drugs used in Cardiac Arrhythmias
- Lipid lowering agents

Hypertension & antihypertensive agents

Hypertension

- Sustained increase in blood pressure $\geq 140/90$
- In general, hypertension is diagnosed if, on two visits, on different days

Classification	Systolic(mmHg)	Diastolic (mmHg)
Normal*	<120	<80
Prehypertension	120-139	80-89
Stage 1	140-159	90-99
Stage 2	>160	>100

*Arterial pressure less than 90/60 mmHg are considered hypotension, and therefore not normal.

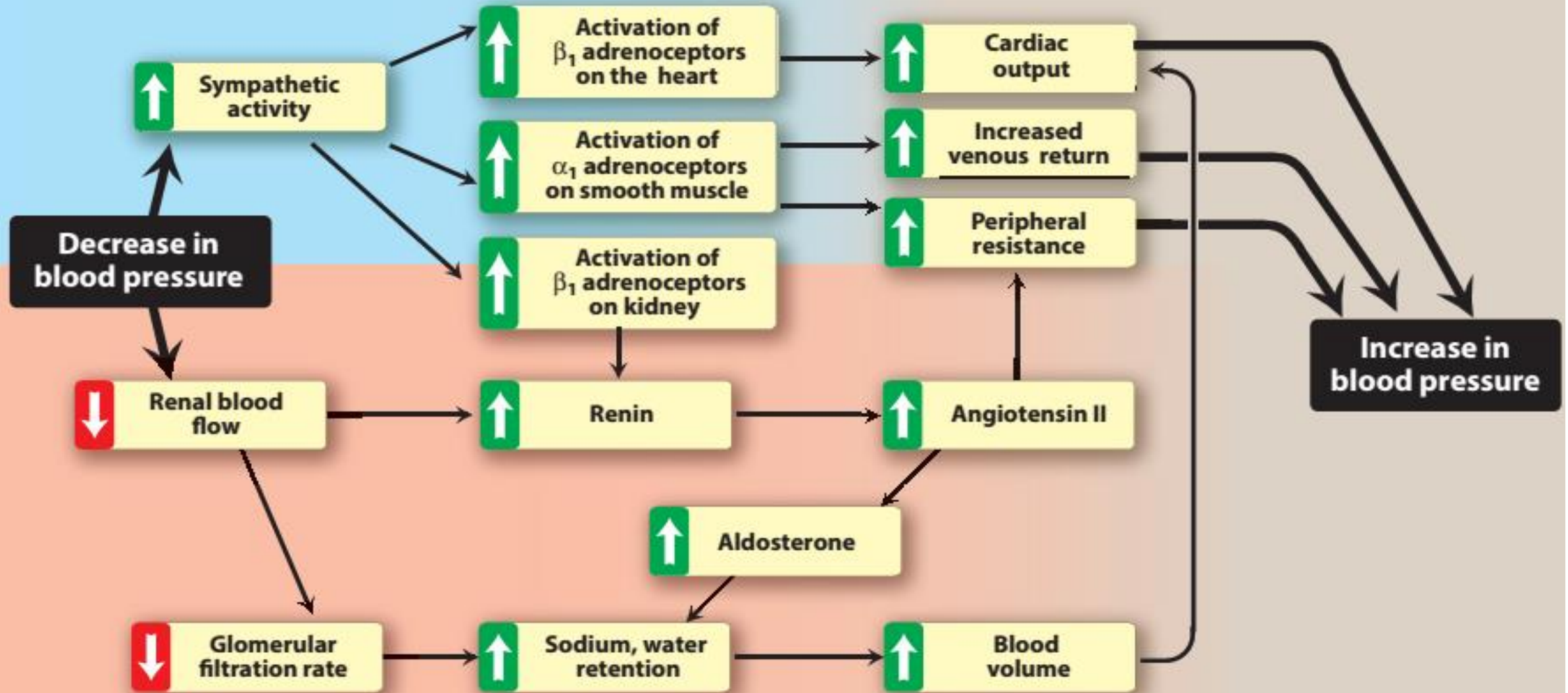
Regulation of BP

- Bp is regulated by short and long term mechanisms
 - **Short term**- sympathetic and parasympathetic activation
 - **Long term**- RAAS
- **Note**: Blood pressure in a hypertensive patient is controlled by the same mechanisms that are operative in normotensive subjects

Long term Renin-Angiotensin-Aldosterone system

- The RAAS is pivotal in long-term BP control
- The RAAS is responsible for:
 - Maintenance of sodium balance
 - Control of blood volume
 - Control of blood pressure
- RAAS (renin release from the juxtaglomerular apparatus) is stimulated by:
 - Fall in BP
 - Fall in circulating volume
 - Sodium depletion

Response mediated by the sympathetic nervous system



Response mediated by the renin-angiotensin-aldosterone system

Fig: Response of the ANS and the renin-angiotensin-aldosterone system to a decrease in blood pressure.

Hypertension

- Two basic types of hypertension
 - Primary (essential) hypertension (85-90%)
 - Treated with drugs in addition to lifestyle changes (e.g., exercise, proper nutrition, weight reduction, stress reduction).
 - Secondary hypertension (10-15%)
 - Identifiable underlying condition such as **renal artery disease, thyroid disease, primary hyperaldosteronism, pregnancy, etc.**

Likely causes essential hypertension

- **↑ reactivity of resistance vessels & resultant ↑ in peripheral resistance**
 - hereditary defect
- **Sodium homeostatic effect**
 - The kidneys are unable to excrete appropriate amounts of sodium for any given BP. As a result sodium and fluid are retained and the BP increases

Other factors

- **Age**-BP tends to rise with age, possibly as a result of decreased arterial compliance
 - Genetics and family history-history of hypertension correlation exists between sibs rather than parent and child
- **Environment**-Mental and physical stress both increase blood pressure
- **Salt intake**-Reducing salt intake in hypertensive individuals does lower blood pressure
- Weight
- Alcohol intake
- Race
 - Black populations are genetically found to be salt retainers 10

Why and how to treat

- The reasons for treating hypertension are obvious
 - Reduce cerebrovascular disease by 40-50%
 - Reduce MI by 16-30%
- Sustained arterial hypertension damages blood vessels in kidney, heart, and brain and leads to an increased incidence of renal failure, coronary disease, heart failure, and stroke.
- How do we treat?
 - stepped approach
 - **use low doses of several drugs?**
 - This approach minimises adverse events and maximises patient compliance

Lifestyle modification (Non pharmacological)

1. Reduced dietary intake of Na^+ and fat,
2. Weight reduction for overweight patients.
3. Regular physical exercise.
4. Stopping smoking and reducing alcohol intake
 - on the average, only modest reductions (5 to 10 mmHg) in blood pressure can be achieved. This may be sufficient for the treatment of some mild hypertensive cases.

Monotherapy Vs Polypharmacy

- Monotherapy of hypertension (treatment with a single drug) is desirable
 - compliance is likely to be better
 - cost is lower
 - Have fewer adverse effects.
- Polypharmacy is common in patients with hypertension require two or more drugs, each acting by different mechanism.
 - Rationale maintaining of blood pressure

Anti-hypertensive drugs

1. Sympatholytics - Which lower blood pressure by reducing peripheral vascular resistance, inhibiting cardiac function, and increasing venous pooling in capacitance vessels.

- Centrally-acting alpha-2 Agonists
- Alpha-1 blockers
- Beta- Blockers

2. Drug acting on Renin-angiotensin-aldosterone system - and thereby reduce peripheral vascular resistance and(potentially) blood volume.

- Angiotensin converting enzyme inhibitor
- Angiotensin II receptor antagonist

3. Diuretics - which lower blood pressure by depleting the body of sodium and reducing blood volume and perhaps by other mechanisms

4. Drug acting on blood vessels - which reduce pressure by relaxing vascular smooth muscle, thus dilating resistance vessels and—to varying degrees—increasing capacitance as well

a. Arteriolar vasodilators

- i. Calcium channel blocker: Verapamil, Diltiazam, Nifedipine
- ii. Hydralazine
- iii. Minoxidil

b. Arteriolar venular vasodilators

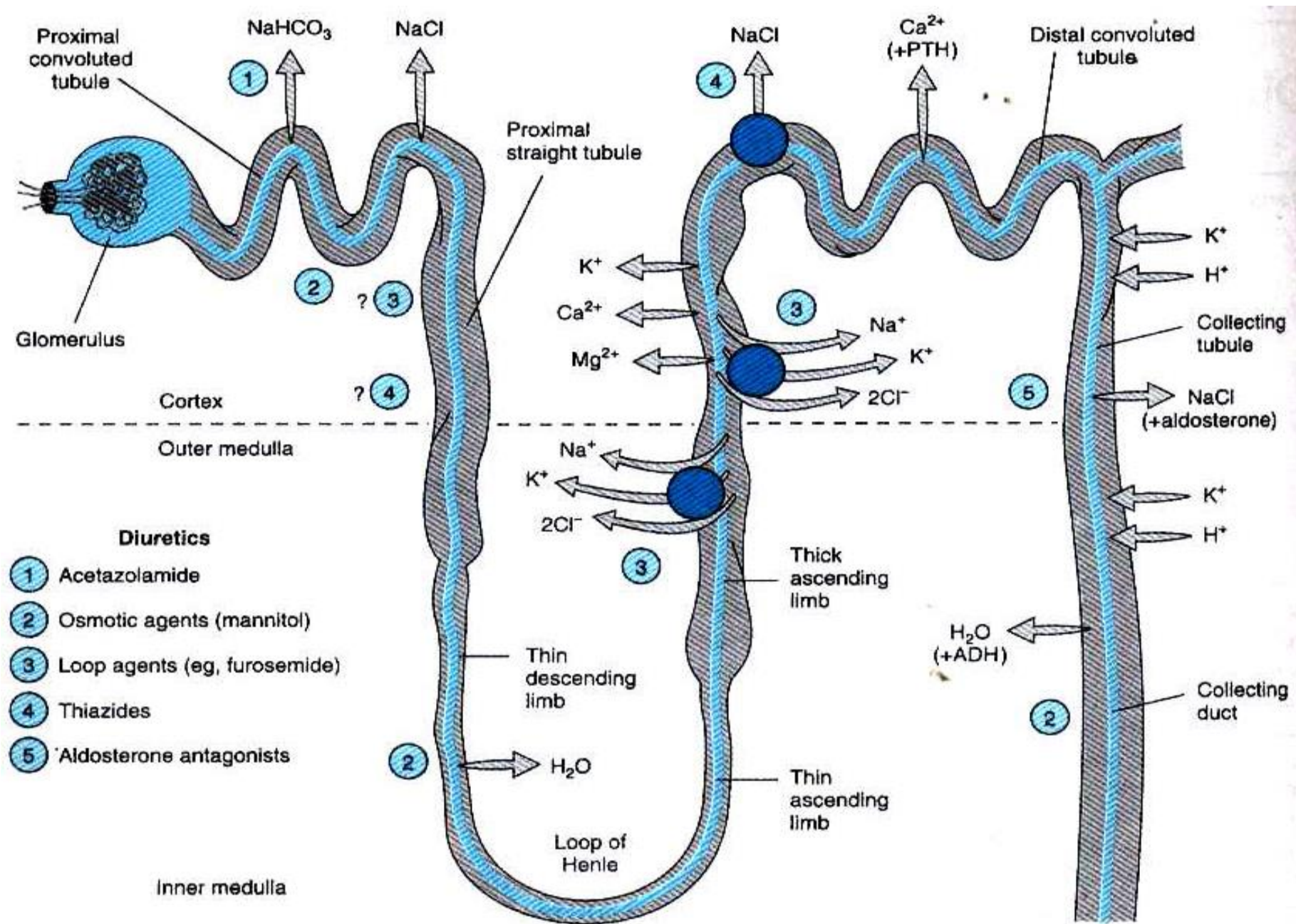
E.g. Sodium nitroprusside

Diuretics

- a “diuretic” is an agent that increases urine volume,
 - Diuretics lower blood pressure primarily by depleting body sodium stores.
 - Diuretics reduce blood pressure by reducing blood volume and cardiac output;
 - Diuretics are effective in lowering blood pressure by 10–15mm Hg in most patients
1. Thiazides and related drugs, e.g. hydrochlorthiazide bendrofluazide,
 2. Loop diuretics, e.g. furosemide, ethacrynic acid, etc.
 3. Potassium sparing diuretics, e.g. spironolactone

Diuretics

- The kidney regulates the ionic composition and volume of urine by the reabsorption or secretion of organic solutes and electrolytes
- at five functional zones along the nephron,
 - Proximal convoluted tubule
 - Descending loop of Henle
 - Ascending loop of Henle
 - Distal convoluted tubule
 - Collecting duct



1.PROXIMAL TUBULE

- Approximately 66% of filtered sodium ions (Na^+), 85% of the NaHCO_3 , 65% of the K^+ , 60% of the water, and virtually all of the filtered glucose and amino acids are reabsorbed in the proximal tubule.
- Of the currently available diuretics, only one group (carbonic anhydrase inhibitors, which block NaHCO_3 reabsorption) acts predominantly in the PCT
- Caffeine is a weak diuretic because it nonspecifically and weakly blocks adenosine receptors that participate in the control of proximal tubule Na^+ reabsorption in the kidney.

A. CARBONIC ANHYDRASE INHIBITORS

- By blocking carbonic anhydrase, inhibitors block NaHCO_3 reabsorption at PCT and cause diuresis.
- The prototypical carbonic anhydrase inhibitor is **acetazolamide**.
- The ciliary body of the eye secretes HCO_3^- from the blood into the aqueous humor.
- Clinical use – Glaucoma - The reduction of aqueous humor formation by carbonic anhydrase inhibitors decreases the intraocular pressure
- Side effects - Hyperchloremic Metabolic Acidosis, renal stone ,

2-Descending loop of Henle

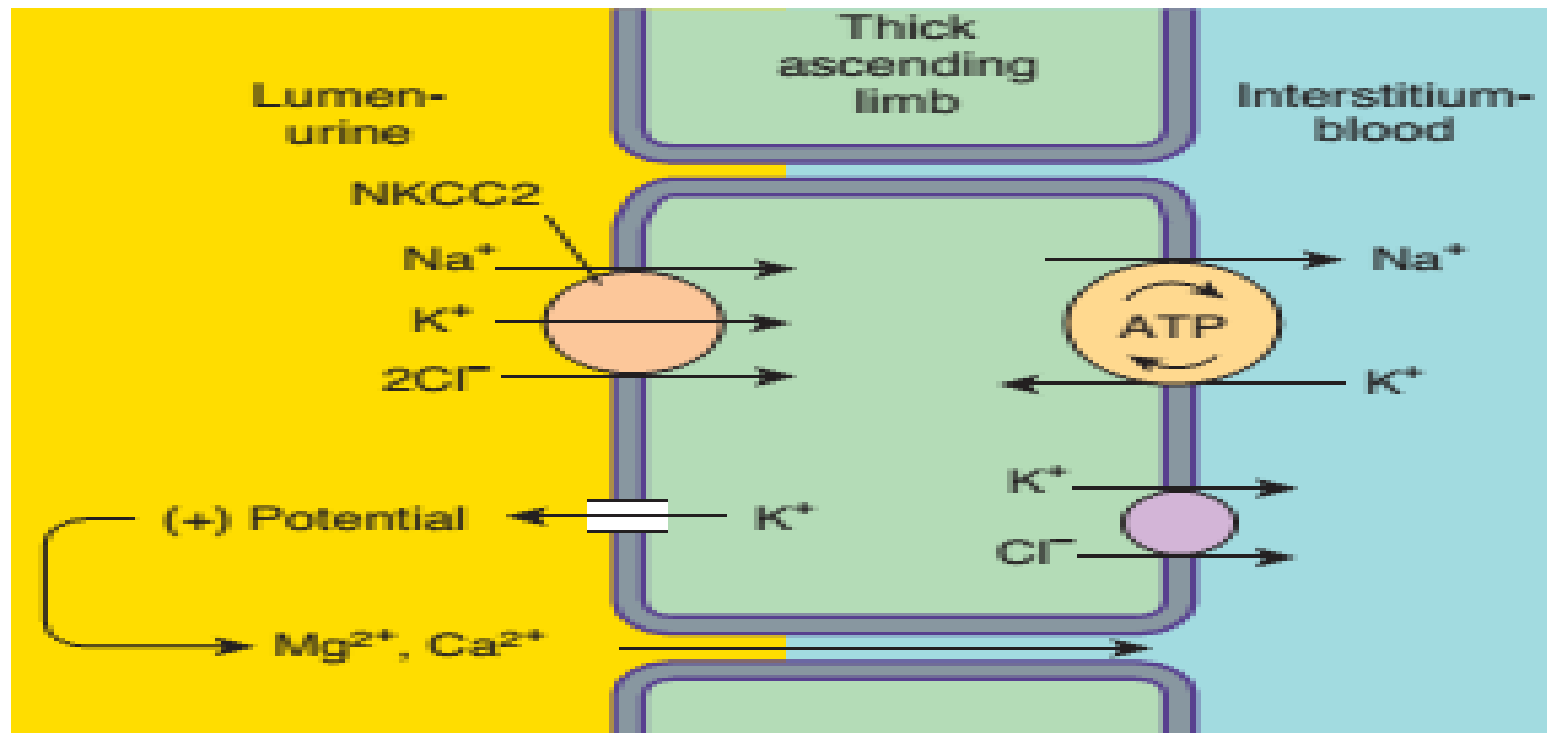
- Permeable only for water
- Results in a three-fold increase in salt concentration.

3-Ascending loop of Henle

- Impermeable to H_2O . (unlike PCT and DLH)
- actively reabsorbs NaCl from the lumen (about 25% of the filtered sodium)
- Active reabsorption of Na^+ , K^+ , and Cl^- is mediated by a $Na^+ / K^+ / 2Cl^-$ co transporter.
- Mg^{2+} and Ca^{2+} enter the interstitial fluid via the paracellular pathway.
- The ascending loop is a diluting region of the nephron.

B- Loop Diuretics

- Furosemide, Bumetanide, Torsemide, Ethacrynic acid
- Selectively block $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ -cotransporter
- Loop diuretics selectively inhibit NaCl reabsorption in the TAL
- loop diuretics are the most efficacious diuretic agents currently available.



Furosemide

20-80 mg

Mechanism of action

- Bind to $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ cotransporter complex
- Site- luminal border of the thick ascending limb of loop of Henle and inhibit Cl^- reabsorption.
- Loop diuretics are more potent than thiazides as diuretics.
- The antihypertensive effect is mainly due to reduction of blood volume.

Pharmacokinetics

- Diuresis begins 30 mins after an oral dose and last up to 6 hrs.
- They act after they are secreted into the kidney tubule by the proximal tubule anion transport mechanisms.

Clinical indications

- The most important indications for the use of the loop diuretics include **acute pulmonary edema**
- Loop diuretics are indicated in cases of severe hypertension which is associated with renal failure, heart failure or liver cirrhosis.
- Acute pulmonary edema of congestive heart failure.
- Acute renal failure.
- Marked edema due to congestive heart failure, nephrotic syndrome, and liver cirrhosis.
- Hypercalcemia

Adverse effects

- Ototoxicity: Hearing can be affected especially if used in conjunction with aminoglycosides. Vestibular function is less likely to be affected.
- Acute hypovolemia, hypotension, and cardiac arrhythmia.
- Hypokalemia and alkalosis.
- Hypomagnesemia predisposing to arrhythmia.
- Hypersensitivity reactions.

4-Distal convoluted tubule (DCT)

- About 10% sodium chloride is reabsorbed via Na⁺/Cl⁻ transporter
- Like the TAL of Henle's loop, this segment is relatively impermeable to water and NaCl reabsorption further dilutes the tubular fluid
- Ca²⁺ excretion is regulated by parathyroid hormone

Thiazide and Thiazide-like diuretics

- **Hydrochlorothiazide**
- Bendroflumethiazide
- Chlorthalidone
- Indapamide.

C. THIAZIDE DIURETICS

The prototypical thiazide is hydrochlorothiazide

Mechanism of action and effects:

- inhibit the Na^+/Cl^- cotransporter
- Site- luminal surface of DCT and early collecting duct
- Initially, thiazide diuretics reduce blood pressure by reducing blood volume and cardiac output as a result of a pronounced increase in urinary water and electrolyte particularly sodium excretion.
- With chronic administration (6-8weeks), they decrease blood pressure by decreasing peripheral vascular resistance as the cardiac output and blood volume return gradually to normal values.

Pharmacokinetics

- All thiazides can be administered orally
- well absorbed from the gut and extensively metabolized in the liver.
- They are mainly secreted via the PCT anion transport mechanism
- Has longer duration than loop diuretics.

Clinical indications

1- Hypertension.

2- Congestive heart failure

3- Renal stones due to idiopathic hypercalciuria

4- Nephrogenic diabetes insipidus.

- Thiazides are appropriate for most patients with mild or moderate hypertension and normal renal and cardiac function.

Contraindication

- hepatic cirrhosis
- borderline renal failure
- congestive heart failure

SIDE EFFECT

- Hypokalemia and alkalosis
- Hyperlipidemia
- Hyponatremia

5-Collecting tubule and duct:

- The cortical collecting tubule (CCT) is responsible for only 2–5% of NaCl reabsorption by the kidney.
- As the final site of NaCl reabsorption, the collecting tubule is responsible for tight regulation of body fluid volume and for determining the final Na⁺ concentration of the urine.
- the collecting tubule is a site at which mineralocorticoids exert a significant influence.
- the collecting tubule is the most important site of K⁺ secretion by the kidney and the site at which virtually all diuretic-induced changes in K⁺ balance occur.
- The collecting tubule is also the site at which the final urine concentration is determined.
- Antidiuretic hormone (ADH, also called arginine vasopressin, AVP) controls the permeability of this segment to water by regulating the insertion of preformed water channels

- Responsible for Na^+ , K^+ exchanges, H^+ secretion, and K^+ reabsorption.
- Stimulation of aldosterone receptors results in Na^+ reabsorption and K^+ secretion.
- Antidiuretic hormone (ADH) receptor promote the reabsorption of water from the collecting tubules and ducts.
- In the absence of ADH, the collecting tubule (and duct) is impermeable to water, and dilute urine is produced. ADH markedly increases water permeability, and this leads to the formation of a more concentrated final urine.

D-Potassium sparing diuretics

- **Spironolactone**
- **Eplerenone**
- **Amiloride**
- **Triamterene**

Mechanism of action

- Site- **late DCT and cortical collecting duct.**
- Potassium-sparing diuretics prevent K⁺collecting tubules
- secretion by antagonizing the effects of aldosterone at the late distal and collecting tubules
- **Spironolactone, canrenone**(active metabolite) and **“eplerenone”** compete with aldosterone for its cytoplasmic receptors.
 - prevent aldosterone-induced transcription and gene expression.
 - work in the presence of aldosterone and their effect is enhanced in hyperaldosteronism.

Clinical indications

- 1- Primary hyperaldosteronism
- 2- Secondary hyperaldosteronism caused by congestive heart failure, hepatic cirrhosis, and nephrotic syndrome.
- 3- Combined with thiazides or loop diuretics.
 - They are used as adjuncts with thiazides or loop diuretics to avoid excessive potassium depletion and to enhance the natriuretic effect of others.
 - The diuretic action of these drugs is weak when administered alone.

Adverse effects:

- 1- **Hyperkalemia** especially in presence of renal disease or other drugs that reduce renin angiotensin system as β -blockers, NSAIDs, ACEIs, or ARBs.
 - Unlike most other diuretics, K^+ -sparing diuretics reduce urinary excretion of K^+
- 2- **metabolic acidosis** due to inhibition of H^+ secretion in parallel with K^+ secretion.
- 3- **Gynecomastia**, impotence, benign prostatic hyperplasia are reported with spironolactone.



Major clinical uses of diuretics

- 1) Edema due to heart failure, nephrotic syndrome, and hepatic cirrhosis.
 - Mild edema can be controlled by thiazide diuretic.
 - More marked edema usually requires the use of a loop diuretic.
 - If resistant, combination
 - If there is marked secondary hyperaldosteronism, spironolactone

2- Hypertension: Low doses of a thiazide diuretic are usually used.

- Loop or spironolactone useful for resistant hypertension.
- Initial hypotensive effects of diuretics is associated with a reduction of blood volume and cardiac output → but after 6-8 weeks blood volume normalizes but peripheral vascular resistance decreases (hence used as an anti-hypertensive)
- Gradual loss of Na^+ → fall in ICF smooth muscle Na^+ → fall in ICF muscle Ca^{2+} → vascular tone decreased

Major.....

- 3- Acute renal failure: A loop or an osmotic diuretic.
- 4- Hypercalcemia: Loop diuretic with saline infusion.
- 5- Hypercalciuria: Thiazide diuretic.
- 6- Glaucoma: acetazolamide and dorzolamide.
- 7- Acute Brain injury: **Mannitol.**

- Thiazides are the preferred diuretics for hypertension
 - single daily dose
 - cause persistent volume depletion which is required to lower BP
- Frusemide
 - once daily dose of frusemide is inadequate as it causes temporary Na^+ loss (BID/TID).
- Thiazides tend to retain Ca^{2+}
 - ↓ risk of bone fracture in the elderly.
- Thiazide diuretics → good for females with osteoporosis-retain Ca^+
- Loop diuretics for severe and in renal insufficiency

Sympatholytics

- All of the agents that lower blood pressure by altering sympathetic function can elicit compensatory effects through mechanisms that are not dependent on adrenergic nerves.
- Thus, the antihypertensive effect of any of these agents used alone may be limited by retention of sodium by the kidney and expansion of blood volume.
- For this reason, sympatholytic antihypertensive drugs are most effective when used concomitantly with a diuretic

Sympatholytics used in Hypertension

Centrally-Acting Drugs			α_1 -receptor Blockers
Imidazoline receptor Agonists-Relmenidine, Moxonidine	α_2 -Receptor agonists Methyldopa	Both Imidazoline & α_2 -Agonists- Clonidine	● Prazosin ● Terazosin ● Doxazosin
Indications			
HTN with diabetes and dyslipidemia: Improve insulin sensitivity & lipid profile.	Hypertension with pregnancy(DOC due to large established safety to fetus).	Severe hypertension (induces rapid reduction of BP).	Hypertension in patients with prostate hypertrophy.
Adverse Effects			
No sedation or dryness	Sedation, dryness, Hepatotoxicity, Hemolytic anemia	Sedation, Dryness, Bradycardia, Rebound $\uparrow$ BP on withdrawal	Posturalhypotension & reflex tachycardia

Centrally-acting alpha-2 Agonists

- Centrally acting sympathetic depressants act by stimulating α_2 -receptors located in the vasomotor centre of the medulla.
- As a result, sympathetic out flow from the medulla is diminished and either total peripheral resistance or cardiac output decreases.
- Drugs that lower blood pressure by actions on the central nervous system tend to cause sedation and mental depression and may produce disturbances of sleep, including nightmares.

Clonidine

- Blood pressure lowering by clonidine results from reduction of cardiac output due to decreased heart rate and relaxation of capacitance vessels, with a reduction in peripheral vascular resistance.

Therapeutic use:

1. Management of mild to moderate hypertension (alone or in combination)
2. Menopausal flushing and prophylaxis of migraine headache
3. **Opiates, alcohol, nicotine withdrawal symptom**
 - Oral dose should be discontinued gradually over 2-4 days
 - Avoid rebound hypertension (receptor upregulation).
 - 2nd choice drug

Adverse effects

- **Dry mouth** and **sedation** are common.
- **Clonidine withdrawal can cause hypertensive crisis (receptors upregulated)**
mediated by increased sympathetic nervous activity.
- Patients exhibit nervousness, tachycardia, headache, and sweating after omitting one or two doses of the drug.
- Depression
 - Avoid alcohol & CNS depressants, gradual withdrawal to prevent rebound hypertension.
- Toxicity: Sedation and drowsiness (20%), sexual dysfunction and xerostomia.

α – Methyldopa.....

Therapeutic use

- Moderate to severe hypertension
- **Methyl dopa is preferably used during pregnancy.**
- The most common undesirable effect of methyldopa is sedation, particularly at the onset of treatment.

Adverse effect

- Headache
- Fatigue
- forget fullness
- Mental depression
- sleep disturbance

α_1 Blockers- Prazosin, terazosin

MOA: decreases TPR and lowers BP by selectively blocking the α_1 adrenergic receptors in vascular smooth muscle (both in the artery and veins).

Effects: HR (unchanged or increased due to reflex tachycardia)

Uses: hypertension, urinary flow obstruction in BPH.

Adverse Effects: severe orthostasis, especially first dose – give at bedtime.

- 2nd choice in mild-moderate hypertension (often used in combination)

β 1-blockers

- β -blockers bind to β -adrenoceptors located in cardiac nodal tissue, the conducting system, and contracting myocytes.
- The heart has both β_1 (predominant) and β_2 adrenoceptors.
- **MOA**- inhibits normal sympathetic effects
 - decrease in cardiac output (β_1 block) and inhibition of renin release (β_1 block of JG cells)
- Propranolol (and other beta blockers); decrease HR, decrease CO, TPR (early increase, later decrease)

β -blockers

- **First generation** β -blockers were non-selective
- **Second generation** β -blockers are more cardioselective (β_1 adrenoceptors).
 - Note that this relative selectivity can be lost at higher drug doses.
- **Third generation** β -blockers are drugs that also possess vasodilator actions through blockade of vascular α -adrenoceptors.
 - **Atenolol, metoprolol**

Beta Blocker	Cardiac Selectivity		Dosing Frequency	Lipid Solubility	$\alpha + \beta$
Acebutolol	++		2	Moderate	-
Atenolol	++		1	Low	-
Betaxolol	++		1	Low	-
Bisoprolol	++		1	Low	-
Carteolol	-		1	Low	-
Carvedilol	-		2	High	+
Esmolol	+		i.v.	Moderate	-
Labetalol	-		2	Moderate	+
Metoprolol	+		1 or 2	Mod. / High	-
Nadolol	-		1	Low	-
Penbutol	-		1	High	-
Pindolol			2	Moderate	-
Propranolol	-		2	High	-
Timolol	-		2	Low / Mod.	-

Sewasew Amsalu (MD)

CVS

Pharmacology

Labetolol (alpha-1,beta blockers);

- Venous tone and TPR decreased, postural hypotension is evident.
- Highly effective when used in hypertensive emergency
- Also carvidolol acts like this

Classic indications

- Hypertension
- Angina
- Myocardial infarction
- Arrhythmias
- Late stages of CHF

Adverse effects

- Bradycardia
- reduced exercise capacity
- heart failure
- hypotension
- atrioventricular (AV) nodal conduction block
- Hypoglycemia
 - Mask symptoms of hypoglycemia

Contraindications

- CHF???????
- COPD, Asthma(non selective)
- heart block >1st degree.
- At present carvedilol and metoprolol have been approved by the FDA for late stages of CHF.
- **Caution:** Diabetes

IV. Vasodilators

- Hydralazine, Minoxidil,
Diazoxide → arterioles
- Nitroprusside → arterioles and
veins
- Calcium channel blockers

Arterial vasodilators

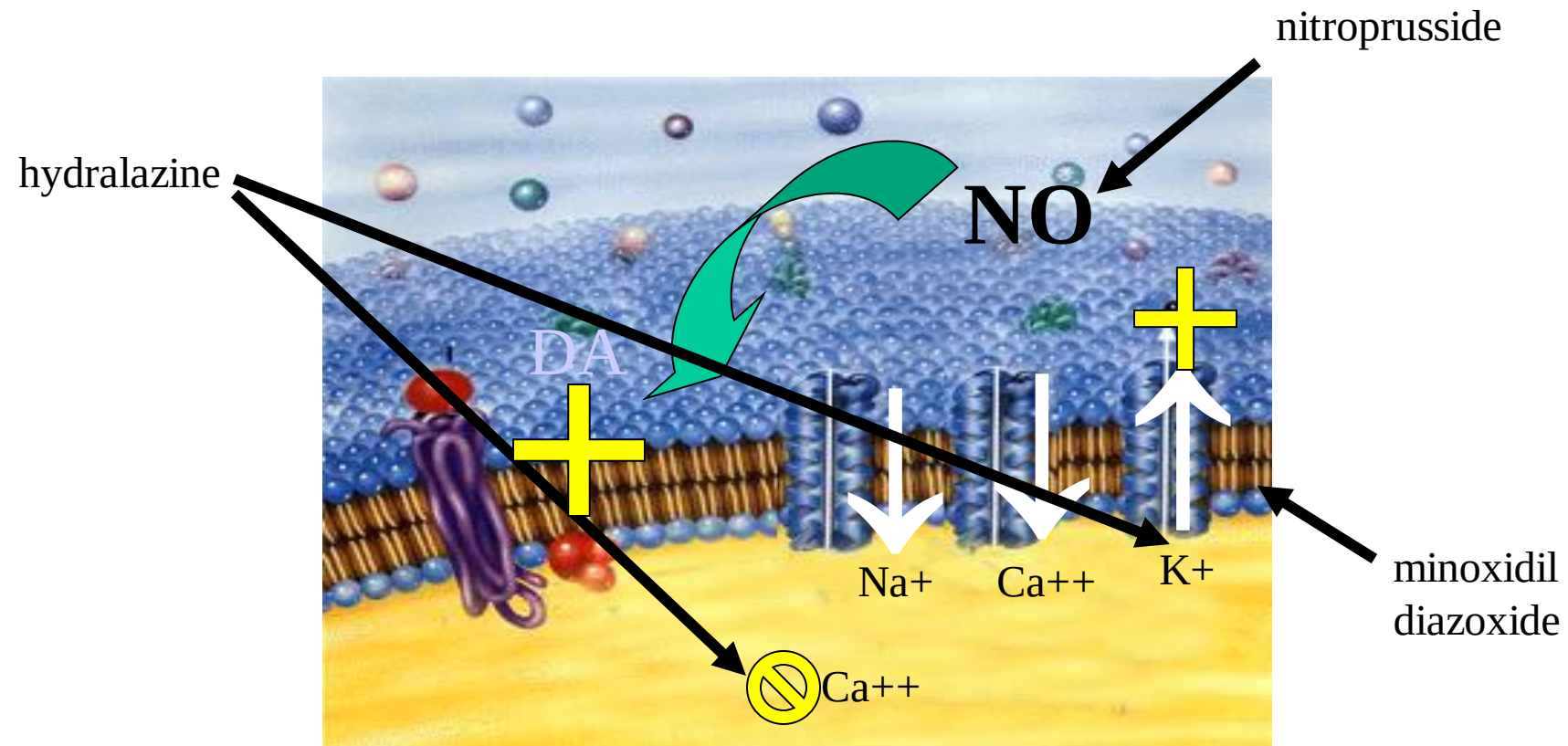
- CCBs
- Hydralazine
- Minoxidil

Artery and vein dilators

- Nitroprusside

Vasodilators

1. Site of Action- vascular smooth muscle



Hydralazine

Mechanism of Action

- Arteriolar vasodilator
- It dilates arterioles but not veins
- It is used particularly in severe hypertension
- K⁺ channel opener
 - hyperpolarization of vascular membrane
 - prevents Ca²⁺ influx into the wall of blood vessels.

Indications

1. Hypertension- IV hydralazine is the DOC in severe hypertension during pregnancy.

2. Congestive Heart Failure

- Not used alone but usually combined with nitrates.
- Potentiates effect of nitrates
 - decreasing after load, nitrate tolerance and free radical formation.

3. Mitral Regurgitation

Adverse Effects

- The most common adverse effects are headache, nausea, anorexia, palpitations, sweating and flushing which are typical to vasodilators.
- Salt retention and edema- need diuretics.
- Reflex tachycardia- need B- blockers.
- Lupus syndrome
- can cause myocardial ischemia (don't use in IHD, CVA)
- **used as a 2nd choice**

Sodium Nitroprusside

Mechanism of Action

- Arteriolar and venular vasodilator
- Donor of NO that increases the level of cGMP which induces vasodilation by inhibiting Ca^{2+} influx into the wall of Blood vessel

Pharmacological Properties

- Potent direct vasodilator (arteriolar and venular) effect decreasing both **preload and afterload**. I.e resulting in reduced peripheral vascular resistance and venous return.
- Nitroprusside rapidly lowers blood pressure and it is given by intravenous infusion
- Has an immediate effect and very **short DOA**(2 minutes).

Indications :-

1. Hypertensive Emergencies

- hypertensive encephalopathy, severe hypertension

2. Severe Acute Heart Failure

- severe acute HF especially mitral and aortic regurgitation.
- myocardial infarction
- cardiac surgery
- Nitroprusside is now replaced by safer drugs as **nitroglycerin or milrinone** (an inotropodilator).
- The most serious toxicities include metabolic acidosis, arrhythmias, excessive hypotension and death.

Minoxidil

MOA: opens K⁺ channels in smooth muscles

membrane stabilizes → contraction less likely

- TPR decreased; HR, CO and renal blood flow is increased
- 3rd or 4th choice drug → many side effects

Adverse effects

- H₂O/Na⁺ retention, edema
- reflex tachycardia, hypertrichosis

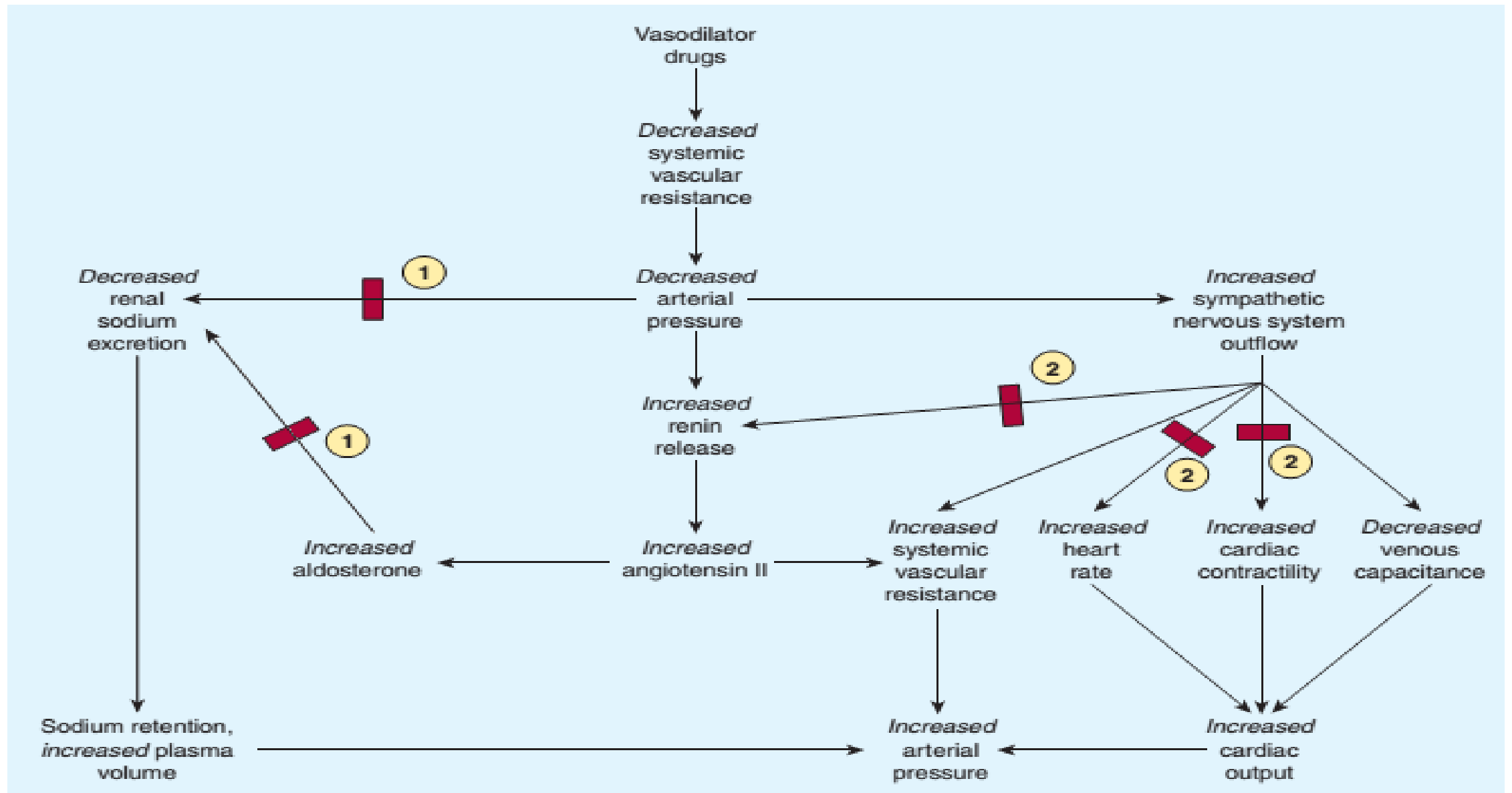


Fig: Compensatory responses to vasodilators; basis for combination therapy with β blockers and diuretics. ¹Effect blocked by diuretics. ²Effect blocked by β blockers

V. Calcium Channel Blockers

Chemical Type	Chemical Names
Non-Dihydropyridines	Verapamil Diltiazem
1,4-Dihydropyridines	Nifedipine Amlodipine

- Non- dihydropyridines – have receptor @ heart
- Dihydropyridines have receptor @arteries

Mechanisms of Action

- Selectively block Ca ions from crossing cell membranes in **cardiac and vascular smooth muscles** without affecting serum Ca levels
- ↓ TPR
- Significant reduction in **afterload** but not preload
- Diuretic action secondary to ↑ **renal blood flow**.
 - ↓ Aldosterone secretion.

Dihydropyridine CCBs

Therapeutic uses

- Hypertension
 - Peripheral vascular disease
 - ❖ More selective for Vessels than cardiac muscles and nodal tissues
- Amlodipine
 - Nifedipine
 - Nicardipine

Non-DHP CCBs

Therapeutic uses

- SV arrhythmia
 - Angina pectoris
 - Hypertension
 - ❖ Selective for nodal tissue and cardiac muscles than blood vessels
- Verapamil
 - Diltiazam

Comparative Adverse Effects

Adverse effect	Verapamil	Diltiazem	Dihydropyridines
Overall	0-3%	10-14%	9-39%
Hypotension	++	++	+++
Headaches	0	+	+++
Peripheral edema	++	++	+++
Constipation	0	++	0
CHF (Worsen)	0	+	0
Caution w/ β blockers	+	++	0
AV block	+	++	0

Angiotensin Inhibitors

1. Angiotensin converting Enzyme inhibitors

- Captopril, Enalapril, lisinopril

2. Angiotensin receptor Blockers

- Losartan, valsartan, Candesartan, eprosartan, irbesartan, telmisartan

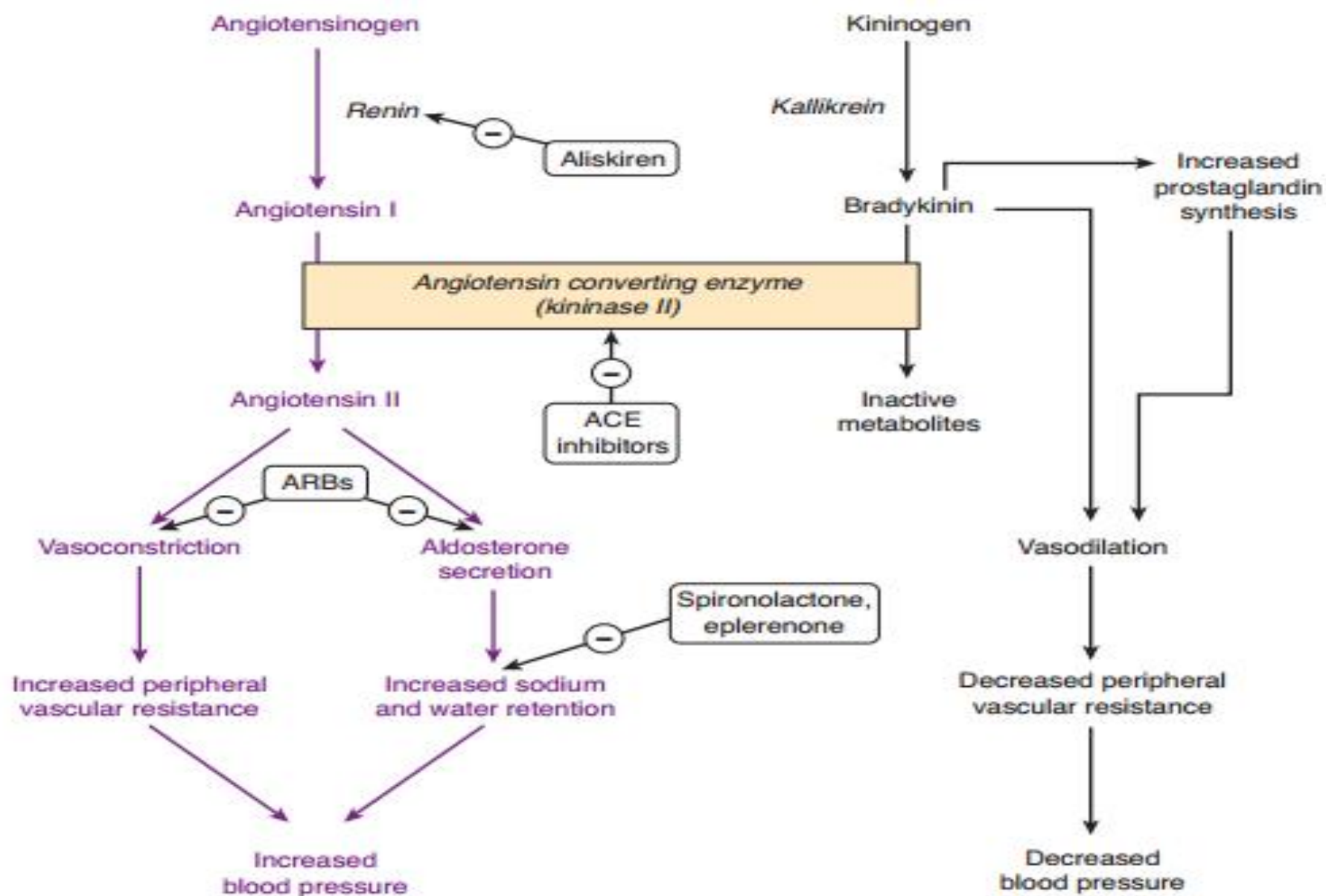


FIGURE 11-5 Sites of action of drugs that interfere with the renin-angiotensin-aldosterone system. ACE, angiotensin-converting enzyme; ARBs, angiotensin receptor blockers.

Renin release

- Stimulated by reduced renal arterial pressure
- Sympathetic neural stimulation
- Reduced sodium delivery
- Increased Na^+ conc. in DCT

ACE inhibitors

- Captopril
- Enalapril
- lisinopril

Class I (Captopril)

- Not a prodrug
- Rapid onset & short duration ($t_{1/2}$ 4-6 h)
- Can be given sublingually in severe hypertension

Class II: (Enalapril)

- Prodrugs (activated first in liver).
- Slow onset & long duration (given once/day).

Class III (lisinopril)

- prodrug
- Long duration
- Water soluble, not metabolized in liver and excreted unchanged by the kidney → given in liver disease

Mechanism of Action of ACEIs

ACEIs have dual vasodilator action by:

- 1. ↓ Angiotensin II formation**
- 2. ↑ Bradykinin through inhibition of its deactivation → direct Vasodilator & release of potent vasodilators**
 - Prostaglandins and NO from vascular endothelium**
Responsible for side effects like dry cough and angioedema

Therapeutic Uses of ACEIs

1. **Hypertension:** ↓ BP, ↓ LV hypertrophy
2. **Ischemic heart disease**
 - inhibits atherogenesis and thrombogenesis.
3. **Myocardial infarction**
4. **Heart failure**
5. **Nephropathy (diabetic or nondiabetic)**

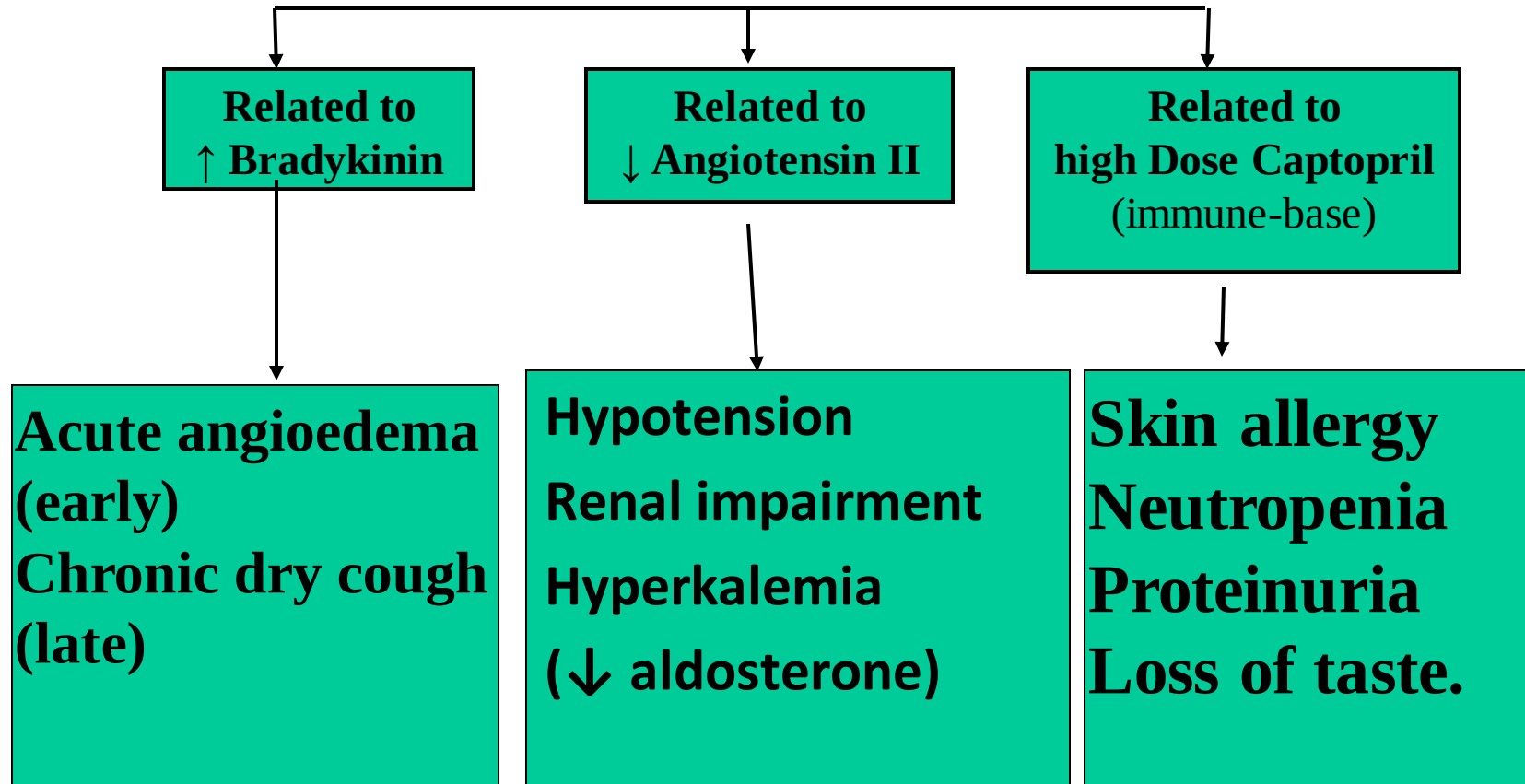
Advantages of ACEIs

1. ↓ Cardiovascular mortality and morbidity.
2. Protect renal function especially in diabetics.
3. No metabolic side effects (no effect on glucose, lipid or uric acid).
4. May improve glucose intolerance in insulin resistance.
5. No changes in heart rate.

Adverse effects

- **Hypotension**
- **Dry cough**
- **Angioedema**
- **First dose syncope and postural hypotension**
- **Hyperkalemia (dm pts, AKI)**
- **Renal insufficiency** (in pts with bilateral renal artery stenosis) – reversible
- **Dysgeusia** (loss of taste)
- **Cough**
- **Angioedema**
- **Potassium change**
- **Taste sensation loss**
- **O hypotension**
- **Pregnancy**(contraindicated)
- **Rash**
- **Increase creatinine** (Renal insufficiency)
- **Lower angiotensin II**

ADVERSE REACTIONS



Contraindications

- Pregnancy-2nd and 3rd trimester
(fetal hypotension, anuria, and renal failure), associated with fetal malformation or death
- Bilateral renal artery stenosis, severe aortic stenosis
- renal insufficiency
- hyperkalemic states
- Coronary artery disease
- Past history of angioedema.

VII. Angiotensin II Receptor Blockers (sartans)

- More selective blockers of angiotensin than ACE
- Have no effect on bradykinin
- Potential for more complete inhibition of angiotensin
- Similar ADR profile except **dry cough and angioedema**
- **Losartan, Valsartan, Irbesartan, Candesartan, Eprosartan, Tasosartan, and Telmisartan**

MOA

- Block AT1 responsible for most of the damaging effects of Ag II
- Produces vasodilation and blocks aldosterone secretion (leads to increased water and salt excretion)

Uses: Hypertension

Management of Hypertension

Management of Hypertension has two components :

I. Non-pharmacologic Treatment(Lifestyle Modifications)

-Indicated for all patients with hypertension

II. Pharmacologic Treatment

- Indicated in patients who failed or unlikely to to achieve target Blood pressure with non pharmacologic treatment alone. Non pharmacologic treatment should be continued while the patients are on pharmacologic treatment.

Target Blood pressure with Drugs on Failure of non pharmacologic Treatment		
Patient	BP	
	SBP	DBP
All adult patients	≥ 140 mmHg	≥ 90 mmHg
Age 60 years and older	≥ 150 mmHg	≥ 90 mmHg

I. Lifestyle Modifications to manage Hypertension

Lifestyle Modifications to Manage Hypertension	
Weight reduction	Attain and maintain BMI<25kg/m ²
Dietary salt reduction	<6gNaCl/d
Adapt DASH-type dietary plan	Diet rich in fruits, vegetables, and low fat dairy products with reduced content of saturated and total fat and reduced salt
Moderation of alcohol consumption	For those who drink alcohol, consume ≤2 drinks/day in men and ≤1 drink/day in women
Physical activity	Regular aerobic activity, e.g., brisk walking for 30 min/d
Cessation of tobacco use	Support with tobacco cessation
Stress management	Behavioural intervention with stress management
Note: BMI, Body Mass Index; DASH, Dietary Approaches to stop Hypertension (trial)	

Lifestyle Interventions in the Management of Hypertension

<u>Intervention</u>	<u>Possible SBP Effect</u>
Exercise	5-10 mm Hg
Weight reduction	1-2 mm Hg/Kg
Alcohol intake reduction	1 mm Hg/drink/d
Sodium intake reduction	1-3 mm Hg/40 mmol/d

Dietary salt intake

Should not exceed 1 teaspoon full (6g) per day.

Avoid adding salt to food on the table

Processed foods often contain high amounts of salt. E.g. bread, processed meats such as bacon, sausages, cheese, margarine, packet soups, tomato sauce, tomato paste, processed spices and other food additives.

Antihypertensive medication

There are five major classes of antihypertensive agents:

- **A**, Angiotensin Converting Enzyme Inhibitors (ACEIs) and Angiotensin receptor blockers (ARBs);
- **B**, β -blockers (BBs);
- **C**, Calcium Channel Blockers (CCBs);
- **D**, Thiazide or thiazide-like diuretics; and
- **Z**, others (sympatholytics, α adrenergic blockers, centrally acting alpha 2-agonists and direct arterial vasodilators.

This last class contains agents that are rarely used, or are obsolete, and examples are as follows:

- Sympatholytics and alpha adrenergic blockers e.g. methyldopa and prazosin
- Direct arterial vasodilators e.g. hydralazine

First line classes of drugs

- **Angiotensin Converting Enzyme Inhibitors (ACEIs) and Angiotensin receptor blockers (ARBs)**
 - **Calcium Channel Blockers (CCBs)**
 - **Thiazide or thiazide-like diuretic**
-
- Long-acting dihydropyridine calcium channel blocker such as amlodipine as first line drug for the treatment of uncomplicated essential hypertension in our country.
 - It is probably effective for all races; reduces need for monitoring of electrolytes and renal function; avoids need for different treatment for women of childbearing age who may become pregnant.

- Thiazide diuretics such as hydrochlorothiazide to be used as add on when target BP not achieved on long-acting dihydropyridine calcium channel blocker such as amlodipine it is less expensive than other hypertension medications in our setting and are probably effective for all races.
- If a third agent is needed, the alternative class of medication is ACE inhibitors
- Beta-blockers are preferred in young patients, high renin hypertension and patients with tachycardia or angina and hypertension.
- Black patients respond well to diuretics and calcium channel blockers than to beta-blockers and ACE inhibitors.

Choice of therapy in Pregnancy

- High BP in pregnancy may usually be due to pre-existing essential HTN or to pre-eclampsia.
- **Methyldopa** is safe in pregnancy.
- **B-blockers** are effective and safe in the third trimester.
- Modified release preparations of nifedipine are also used in HTN in pregnancy.
- IV labetalol or hydralazine can be used to control hypertensive crisis.
- Magnesium sulphate is the drug of choice to prevent seizures in pre-eclampsia and eclampsia
- Pregnant women and women of childbearing age not on effective contraception should not be given ACE inhibitors, ARBs, or thiazide/thiazide-like diuretics; CCBs should be used.

Drugs used in Heart failure

Heart failure

- **Heart failure**- inability of the heart to keep up its work load of pumping blood to the lungs and to the rest of the body.
- Heart failure occurs when cardiac output is inadequate to provide the oxygen needed by the body

Characterized by:

- Impaired ventricular performance (right or left or both)
- Exercise intolerance: dyspnea, fatigue
- Shortened life expectancy
- Fluid retention

Heart failure cont'd...

- Underlying causes of HF include
 - Arteriosclerotic heart disease
 - **Myocardial infarction (IHD)**
 - **Hypertensive heart disease (HTN)**
 - **Valvular heart disease**
 - Dilated cardiomyopathy, and
 - Congenital heart disease.

Pathogenesis of CHF

- Lack or loss of contractile force $\Rightarrow$ $\downarrow$ ed ventricular function $\Rightarrow$ reduced CO
- As a result a variety of adaptive mechanisms are activated
 - ✓ The compensatory mechanisms are either intrinsic or extrinsic
- The primary signs and symptoms of all types of heart failure include tachycardia, decreased exercise tolerance, shortness of breath, peripheral and pulmonary edema, and cardiomegaly.
- Decreased exercise tolerance with rapid muscular fatigue is the major direct consequence of diminished cardiac output

Neuro-humoral (extrinsic) compensatory mechanisms

I. Sympathetic nervous system over activity

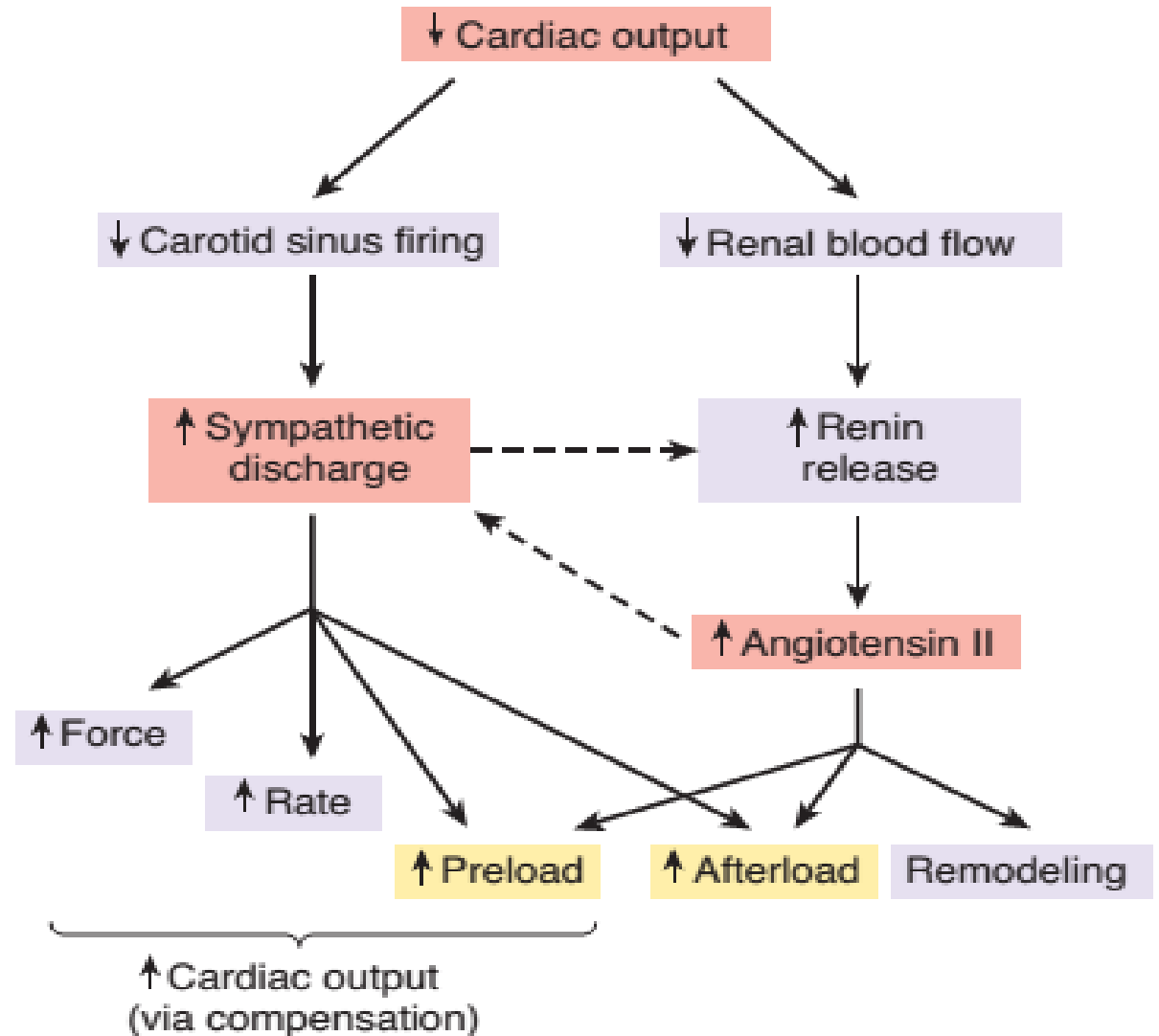
- ✓ ↑ed sympathetic and ↓ed Parasympathetic outflow
- ✓ Initial increase in HR, contractility, and vascular tone
 - Increased preload, force, & HR; increases the CO
 - ↑ed arterial tone ↑es the after load which leads to ↓ed ejection fraction and hence, reduced CO
- ✓ Increased RAAS

II. Increased release of ADH

- ✓ Increased water reabsorption

Intrinsic (cardiac) compensatory mechanisms

- Myocardial hypertrophy
- 'Remodeling'



➤ Pathophysiology of cardiac performance

- Is a function of four primary variables

- ✓ Increased preload

- Rx: reducing preload (salt restriction, diuretic therapy and venodilator drugs)

- ✓ Increased Afterload

- Rx: reducing arterial tone (arteriolar vasodilators)

- ✓ Depressed intrinsic contractility of myocardium

- Rx: increasing contractility using inotropic agents

- ✓ Increased HR due to sympathetic over activity

- Rx: reducing the HR (β blockers)

Congestive Heart Failure

Pathophysiology: Cardiac Output

Determined By Several Factors

- Preload:** (**Atrial Pressure**) Increased in heart failure due to increased blood volume and venous tone **Treated with salt restriction and diuretics**
- Afterload:** (**Vascular Resistance**) Increased due to reflex sympathetic outflow and renin-angiotensin system though elevated **afterload** may further reduce cardiac output **Reduction of arterial tone**
- Contractility:** Reduction in intrinsic contractility and therefore reduction in pump performance **Inotropic drugs to increase contractility**
- Heart Rate:** Increases through sympathetic NS compensation

Drug groups commonly used in Heart Failure

- Inotropic agents
 - Cardiac glycosides
 - β agonists
 - PDE inhibitors
- Diuretics
- ACE inhibitors
- Angiotensin receptor blockers
- Aldosterone antagonists
- Direct vaso- and venodilators
- **β blockers????**

1. Cardiac glycosides.

- Cardiac glycosides comprise a group of steroid compounds that can increase cardiac output and alter the electrical functions.
- Commonly used cardiac glycosides are **digoxin** and **digitoxin**.
- All cardiac glycosides exhibit similar pharmacodynamic properties but do differ in their pharmacokinetic properties.
- For example, digitoxin is more lipid soluble and has long half-life than digoxin.

Digoxin

MOA-

- Binding to $\text{Na}^+/\text{K}^+-\text{ATPase}$ in cardiac cell membrane
 - $\uparrow$ intracellular Na^+
 - $\downarrow$ Ca^{2+} expulsion by $\text{Na}^+/\text{Ca}^{2+}$ exchanger
 - $\uparrow$ intracellular Ca^{2+} that increase force of contraction
- Sensitize $\text{Na}^+/\text{K}^+-\text{ATPase}$ activity in vagal nerves
 - $\downarrow$ heart rate

Clinical indication of digoxin

- **Congestive heart failure**
- **Arrhythmias (atrial fibrillation)**
- **It has narrow therapeutic index: high risk for toxicity**
- Serum level ($< 2-3$ ng/ml) monitoring is recommended

Toxicity of digoxin

Toxicity of cardiac glycosides include:

- Gastrointestinal effects such as anorexia, nausea, vomiting, diarrhoea
- Cardiac effects such as bradycardia, heart block, arrhythmias
- CNS effects such as headache, malaise, hallucinations, delirium, visual disturbances (yellow vision)

Management of digitalis toxicity

- If mild GI or Visual disturbances - reduce the dose
- If cardiac arrhythmias occur check serum levels of K^+ , Digoxin, Ca^{++}
- Correct electrolytes
- ✓ Use anti arrhythmic agents like Lidocaine
- ✓ Administer **digitalis antibodies**. E.g., *digibind*

Beta agonists

❑ Dopamine (i.v.)

- Short duration of action
- Use only low and moderate doses
 - May improve sodium and water excretion in patients refractory to loop diuretics
- High doses are potent vasoconstrictor
- when systolic pressure > 90 mm Hg
 - Add nitroprusside to reduce preload and afterload
 - Add I.V. furosemide to reduce edema

Beta agonists cont'd...

□ Dobutamine

- Selective β_1 agonist with short $t_{1/2}$ (2.4 min)
- IV infusion to treat acute severe heart failure
- After 72 hours of therapy, tolerance can develop
- Can enhance AV conduction and worsen atrial tachycardias

2. Diuretics

- Diuretics are the mainstay of heart failure management
- Diuretics are first – line drugs for treatment of patients with heart failure.
- They have no direct effect on cardiac contractility; their major mechanism of action in heart failure is to reduce venous pressure and ventricular preload.
- This results in reduction of salt and water retention and edema and its symptoms.
- The reduction in venous pressure causes reduction of edema and its symptoms and reduction of cardiac size which leads to improved efficiency of pump function.
- Especially loop diuretics

Angiotensin antagonists

- Synthesis inhibitors: ACEIs
- Receptor antagonists: sartans
- First line agents with diuretics
- These drugs reduce after load by reducing peripheral resistance and also reduce preload by reducing salt and water retention by way of reduction in aldosterone secretion.

ACEIs

- A major breakthrough in the treatment of CHF
- Improve symptoms, reduce mortality, slow down progressive deterioration in cardiac function
- Fluid retention can attenuate the effects of ACEIs, optimize the dose of diuretic first, before ACEI
 - reduce the dose of diuretic during the initiation of an ACEI to prevent symptomatic hypotension

Clinical indications

- CHF -reduce morbidity and mortality
- Post MI - reduce morbidity, mortality and onset of heart failure

Adverse effects- refer hypertension

ARBs

- ARBs selectively block the angiotensin II, AT1 receptor
- They are effective in the treatment of heart failure
- **NOT AS EFFECTIVE AS ACEIs in CHF**
- At present recommended for use in ACEI intolerant patients
- Adverse effects- refer hypertension

Beta blockers(Carvedilol, Metoprolol)

- **Most patients with chronic heart failure respond favorably to certain B-blockers in spite of the fact that these drugs can precipitate acute decompensation of cardiac function**

Metoprolol- 2nd generation 75 fold higher affinity for β_1 than β_2 receptor

- long-acting, controlled release form used for CHF
- Also approved for HTN and IHD

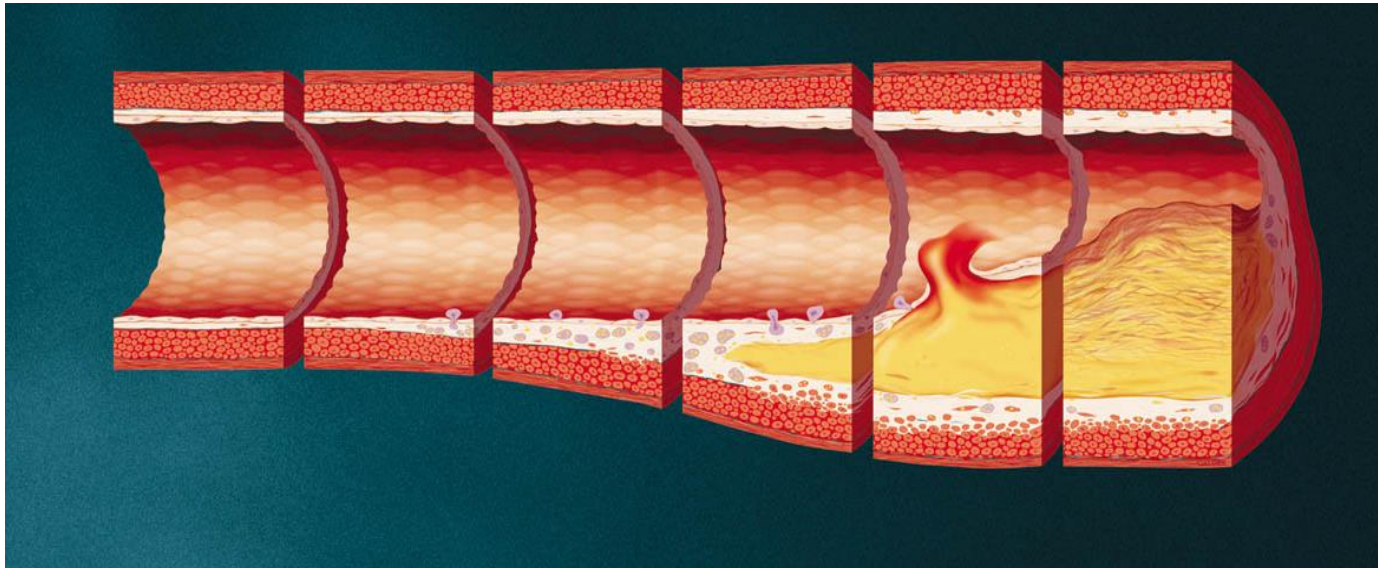
Carevedilol-Approved for CHF in most countries

- Minimally β -1 selective (low dose), high dose has affinity for α_1 as well as antioxidant action
- Orthostatic hypotension more prominent
- High dose block all receptors (α & β)

Diuresis:

- Send sample for Cr, BUN, K⁺ and Na⁺ initially and proceed with diuresis.
- For diuretic naive patients start furosemide 40 mg IV if BP > 90/60 mmHg and double the dose every 2-4 hour until the urine output is > 1 ml/kg/hr (40-70ml/hr). Response to IV dose occurs 2-4 hours later.
- Start one of the ACEIs/ARBs as soon as BP and RFTs permit and escalate until discharge (*see chronic heart failure section*)
- Start one of the BB following ACEIs/ARBs when BP and PR permit and escalate until discharge (*see chronic heart failure section*)
- Start Spironolactone 25mg/d.

Pharmacology of Ischemic Heart Disease



Ischemic Heart Disease

- Myocardial ischemia results from
 - Fixed partial obstruction of an artery or arteries due to atheroma or other pathologies
 - Inappropriate arterial vasospasm
 - Thrombus
- Myocardial ischemia could be
 - Partial or complete
 - Temporary or permanent (death of a tissue by necrosis or apoptosis)
 - Acute: Myocardial Infarction
 - Chronic: Angina pectoris, Silent Ischemia

Ischemic Heart Disease

- If Myocardial ischemia is sufficiently severe and maintained for a long period of time leads to tissue infarction
- Ischemia with pain preferred than silent, as it gives a warning signal for intervention
- Myocytes depend on aerobic metabolism, if oxygen remains low die by apoptosis or necrosis.
- Angina pectoris develops as a result of an imbalance between the oxygen supply and the oxygen demand of the myocardium

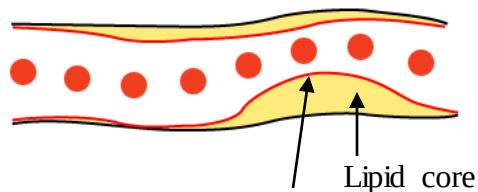
Chronic Ischemic Heart Disease

- Chronic IHD arises as a result of a mismatch between myocardial blood/oxygen supply and demand
- Attacks of angina may be precipitated by any stress which increases cardiac work and myocardial oxygen demand
- The heart extracts approximately 75% of available oxygen even under no stress

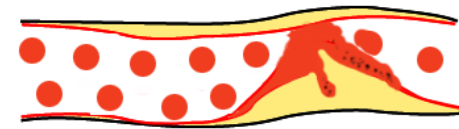
Angina pectoris

- Angina pectoris is the principle symptom of ischemic heart disease
- When the increase in coronary blood flow is unable to match the increased oxygen demand, angina develops.
- It has become apparent that spasm of the coronary arteries is important in the production of angina.
- Characterized by **sudden, severe and pressing substernal pain** that is usually felt beneath the upper sternum and is often transferred to the surface areas of the body and most often to the **left arm** and **shoulder** and even to the side of the **face**

- In classic angina, the imbalance occurs when the myocardial oxygen requirement increases, as during exercise, and coronary blood flow does not increase proportionately.
- The resulting ischemia usually leads to pain.



Plaque rupture



Acute closure of the
artery by a thrombus

❑ Ischemic condition results from

➤ Imbalance b/n myocardial oxygen demand & oxygen supplied by coronary vessels.

- **Increased myocardial oxygen demand**

- ✓ Determined by **Ventricular wall tension**, HR, contractility

- **Decreased myocardial oxygen supply**

- ✓ Determined by **coronary blood flow (CBF)**, oxygen-carrying capacity of the blood

❑ In theory, the imbalance between oxygen delivery and myocardial oxygen demand can be corrected by **decreasing oxygen demand** or by **increasing delivery**(by increasing coronary flow).

Types of Angina

1. Stable /exertional, typical, classic.../ Angina

- **Underlying Pathology:** atherosclerosis in large coronary arteries
- Episodes **precipitated** by exercise, cold, stress, emotion, eating
- **Treatment principles:** Decrease cardiac load (pre-& after load), increase myocardial blood flow
 - ✓ Organic nitrates, β -blockers, CCB, Statins (atehroma), antiplatelet drugs (aspirin)

TYPES....

2. Vasospastic/Variant, Prinzmetal's.../ Angina

- **Cause:** transient Vasospasm of coronary Vessels
- Associated: underlying atheromas
- Pain can occur at rest
- **Treatment principles:** ↓ed Vasospasm of coronary Vessels
 - **Coronary vasodilators(nitroglycerin) and CCBs**

TYPES...

3. Unstable /preinfarction, crescendo.../ Angina

- Chest pains occur with ↑ed frequency and are precipitated by progressively less effort.
- **Cause:** recurrent episodes of small platelet clots.
- Site: ruptured atherosclerotic plaque
- Precipitated: local Vasospasm
- **Treatment principles:** inhibit platelet aggregation & thrombus formation, decrease cardiac load, Vasodilate coronary arteries.

Therapeutic rationale

- Decreasing myocardial oxygen demand:
 - Reducing cardiac workload
 - Reducing heart rate
 - Reducing force of myocardial contraction
 - Reducing after load
- Increasing myocardial oxygen supply:
 - Vasodilators

Classification of Antianginal Agents

1. Organic nitrates

- Reduce preloads & after load, dilate coronary arteries.
- Inhibits platelet aggregation.

2. Ca^{++} channel blockers

- Vasodilate coronary arteries. Reduce after load, inhibit platelet aggregation
- Some; decrease HR, decrease contractility.

3. β -adrenergic antagonists

- Decrease HR, contractility and afterload

Drug Therapy

- **Other drugs**
- Aspirin
- Cholesterol lowering agents
 - HMG CoA reductase inhibitors
- Lipid lowering drugs, especially the “statins,” have become extremely important in the long-term treatment of atherosclerotic disease

1. Nitrovasodilators

- **Nitrates**

- Glyceryl Tri Nitrate (GTN)-sublingual, buccal, transdermal
- Isosorbide Mononitrate- sustained release, tablets
- Isosorbide Dinitrate- sustained release, tablets

- **Nitrites**

- Amyl nitrite
- Isobutyl nitrite

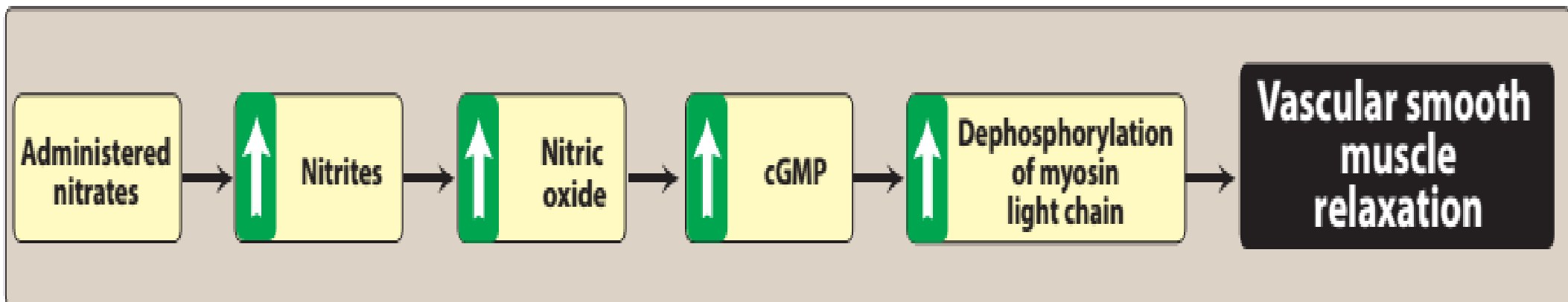
- **NO containing compounds**

- Sodium nitroprusside

- organic nitrates are potent vasodilators and successfully used in therapy of angina pectoris for over 100 years
- **Nitroglycerin** may be considered the prototype of the group
- All therapeutically active agents in the nitrate group have identical mechanisms of action and similar toxicities

Mechanism of action

- The effects of nitrates are mediated through the direct relaxant action on smooth muscles.
- Nitrates are believed to act by mimicking the vasodilator action of endothelium derived relaxing factor (EDRF) identified as nitric oxide.
- Vasodilating organic nitrates are reduced to organic nitrites, which is then converted to nitric oxide.
- Smooth muscle relaxation: due to



Pharmacological effects

- ❑ Venodilation (major ↓ preload)
- ❑ Arteriolar dilatation (↓ afterload)
 - ↓ Myocardial work load → ↓ se in O₂ demand.
- ❑ Improve perfusion of ischemic myocardium.
- ❑ Inhibition of platelet aggregation

Pharmacokinetics

- Oral bioavailability is very low
 - Extensive hepatic first pass metabolism.
 - **Sublingual** effective for the Rx of acute attacks of angina pectoris due to first pass effect
 - Long acting preparations (oral, transdermal, ointment) used to provide prolonged prophylaxis against angina attacks
- The action of nitrates begins after 2-3 minutes when chewed or held under tongue and action lasts for 2 hours.

Organic nitrates...

➤ Therapeutic uses

- Treatment & prevention of all types of Angina
 - ✓ Variant angina
 - ✓ Stable angina
 - ✓ Unstable angina

Drug	Dose	Duration of Action
"Short-acting"		
Nitroglycerin, sublingual	0.15–1.2 mg	10–30 minutes
Isosorbide dinitrate, sublingual	2.5–5 mg	10–60 minutes
Amyl nitrite, inhalant	0.18–0.3 mL	3–5 minutes
"Long-acting"		
Nitroglycerin, oral sustained-action	6.5–13 mg per 6–8 hours	6–8 hours
Nitroglycerin, 2% ointment, transdermal	1–1.5 inches per 4 hours	3–6 hours
Nitroglycerin, slow-release, buccal	1–2 mg per 4 hours	3–6 hours
Nitroglycerin, slow-release patch, transdermal	10–25 mg per 24 hours (one patch per day)	8–10 hours
Isosorbide dinitrate, sublingual	2.5–10 mg per 2 hours	1.5–2 hours
Isosorbide dinitrate, oral	10–60 mg per 4–6 hours	4–6 hours
Isosorbide dinitrate, chewable oral	5–10 mg per 2–4 hours	2–3 hours
Isosorbide mononitrate oral	20 mg per 12 hours	6–10 hours

Tolerance

- Tolerance can develop rapidly
- Magnitude is a function of dosage and frequency

How to overcome tolerance?

Using a sustained release preparation which incorporates a “nitrate free period

Adverse effects

- The major acute toxicities of organic nitrates are direct extensions of therapeutic vasodilation: orthostatic hypotension, tachycardia, and throbbing headache
- Headache, dizziness
 - Increase dose slowly
- Postural hypotension (**GTN syncope**)
- Reflex tachycardia
- Flushing, methemoglobinemia
- **Withdrawal from long-term exposure can produce severe ischemia resulting in death or MI**
- Dangerous hypotension: PDE5 inhibitors (sildenafil) so highly contraindicated

2. Calcium Channel Blockers (CCBs)

- It has been known since the late 1800s that calcium influx is necessary for the contraction of smooth and cardiac muscle
- Increased free calcium in the cytoplasm of vascular smooth muscle cells leads to vasoconstriction
- Calcium antagonists block voltage-dependent channels. These channels allow the entry of calcium in response to cell depolarization.
- Calcium channel blockers appear to involve their interference with the calcium entry into the myocardial and vascular smooth muscle, thus decreasing the availability of the intracellular calcium

CCB...

- Site of action of Ca^{++} -channel blockers.
 - Bind to L-type Ca^{++} -channels- prevent Ca^{++} influx.
 - ✓ different classes bind to different sites on the channel
 - ✓ different types of L-type channels exist.
 - Bepridil; also blocks Na^+ and K^+ channels
 - Mibefradil: also blocks T-type channels

Non-DHP CCBs

Therapeutic uses

- SV arrhythmia
 - Angina pectoris
 - Hypertension
 - ❖ Selective for nodal tissue and cardiac muscles than blood vessels
- Verapamil
 - Diltiazam

Dihydropyridine CCBs

Therapeutic uses

- Hypertension
 - Peripheral vascular disease
 - ❖ More selective for Vessels than cardiac muscles and nodal tissues
- Amlodipine
 - Nifedipine
 - Nicardipine

Pharmacological Effects

➤ Vascular smooth muscle

- Decreased intracellular Ca^{++} in arterial S. muscle → Relaxation

- ✓ Decreased after load

- Little or No effect on venous beds

- ✓ No effect on preload.

➤ Calcium channel blockers decrease myocardial contractile force, which reduces myocardial oxygen requirements.

➤ Calcium channel block in arterial smooth muscle decreases arterial and intraventricular pressure.

- Decreased heart rate with the use of verapamil or diltiazem causes a further decrease in myocardial oxygen demand
 - Calcium channel-blocking agents also relieve and prevent the focal coronary artery spasm involved in variant angina.
-
- **Use of these agents has thus emerged as the most effective prophylactic treatment for this form of angina pectoris.**

CCB..

➤ Adverse drug effects

- Due to excessive Vasodilatation: dizziness, headache, hypotension, flushing, edema etc.
- Aggravation of myocardial ischemia: reflex tachycardia
- Bradycardia, exacerbation of CHF

➤ Contraindications (specially Verapamil & Diltiazem)

- Moderate to severe ventricular dysfunction
- SA or AV conduction disturbances
- Systolic BP less than 90mm Hg

CCB..

➤ Therapeutic Uses

- Angina; Variant, Exertional, and Unstable
- Are also employed in supraventricular arrhythmias and Hypertension

3. Beta blockers

- Although they are not vasodilators B-blocking drugs are extremely useful in the management of effort angina.
- The beneficial effects of B-blocking agents are related primarily to their hemodynamic effects—decreased heart rate, blood pressure, and contractility—which decrease myocardial oxygen requirements at rest and during exercise
- Exercise and emotional excitement induce angina in susceptible subjects by the increase in heart rate, blood pressure and myocardial contractility through increased sympathetic activity.

- Beta receptor blocking agents prevent angina by blocking all these effects.
- Increase in HR often precipitate angina, drug blunting sympathetic effect on the heart would be antianginal
- B-blockers are reversible antagonists of the β_1 and β_2 receptors
 - Cardioselective drugs (β_1) useful in treating exertional angina attacks (metoprolol, atenolol)
 - High dose: selectivity is lost and effect resembles to that of non-selectives
- **Not effective against vasospastic angina due to increased coronary resistance by acting at α adrenergic receptors**

Mechanism of action

- Decrease three major determinants of myocardial oxygen demand
 - Heart rate, Contractility and Systolic wall tension
- In unstable angina, their sympatholytic effects reduce cardiac workload and myocardial oxygen demand
 - 13% relative risk reduction in rate of progression to an acute MI
 - 29% risk reduction in death in high risk individuals with evolving MI
- Beta-blocking agents decrease mortality of patients with recent myocardial infarction and improve survival and prevent stroke in patients with hypertension

➤ Therapeutic Uses

- Exertional angina
- Unstable angina; reduce progression to MI.
- Myocardial infarction: improve mortality
- Combined with Nitrates &/or Ca^{++} channel blockers.

➤ Not useful in Variant Angina

Rebound phenomenon

- Sudden cessation of β -blocker therapy may precipitate myocardial infarction

Adverse effects

- Bradycardia
- reduced exercise capacity
- heart failure
- hypotension
- atrioventricular (AV) nodal conduction block
- Hypoglycemia
 - Mask symptoms of hypoglycemia

Antiplatelet Agents

- Low dose Aspirin (80-325 mg)
- The formation of platelet aggregates are important in the pathogenesis of angina, unstable angina and acute MI
- Aspirin is a potent inhibitor of platelet thromboxane production
- Thromboxane stimulates platelet aggregation and vasoconstriction
- Recommended for stable and unstable angina
- Higher doses may increase bleeding

Therapeutic use

- regular daily use of aspirin
 - In acute MI: reduce mortality by 23% and in combination with streptokinase reduce mortality by 42% and reinfarction by 52%
 - In unstable angina reduce MI and death by 50%
- In secondary prevention
 - reduce reinfarction by 32% and combined vascular events by 25%

Clopidogrel

- Inhibits ADP receptor platelet aggregation
- A 9% reduction in adverse cardiovascular events
- Initiate a loading dose of 300 mg and then 75 mg once daily
- In unstable angina combined with aspirin reduces mortality
- Avoid within first few days of an MI and first 7 days of an ischemic stroke (no clear benefit)
- Same incidence of bleeding as aspirin but possibly lower GI bleeding

Statins

- These are HMG CoA Reductase Inhibitors which are the most effective cholesterol lowering agents
- Standard therapy for most patients with unstable angina
- Start treatment with high dose of atorvastatin (80 mg/d), pravastatin (40 mg/d)
 - Risk reduction >10%
- Studies have demonstrated that aggressive cholesterol lowering post MI can reduce cardiovascular mortality by 42% and total mortality by 30%

- Provide pain control:
 - Nitroglycerin: short acting (sublingual or spray): don't give nitrates for patients who took phosphodiesterase inhibitors (sildenafil)
 - Morphine 4 mg IV if pain refractory to nitrate therapy alone.
- Aggressive medical management is indicated: treat as in MI
 - Dual antiplatelet therapy:
 - ASA 300 mg loading followed by 81-100 mg Po daily **AND**
 - Clopidogrel 300 mg loading followed by 75 mg Po daily
 - β -Blockers:
 - Metoprolol 12.5 mg Po BID and titrate as needed
 - Anticoagulation using Heparin for the first five days until ambulation starts
 - Enoxaparin 1 mg/kg SC BID
 - Alternative: Unfractionated Heparin 5000 IU IV loading followed by 17,500 IU SC BID
 - High intensity statin therapy
 - Atorvastatin 80 mg PO daily

Lipid Lowering Agents

Introduction

➤ Lipid:

- Free fatty acid (FFA)
- Triglycerides (TG)
- Phospholipids
- Cholesterol and cholesterylesters

➤ Lipids are insoluble in water (hence in blood)

➤ Transport:

- FFA $\Rightarrow$ bound to albumin
- TGs, cholesterol, phospholipids $\Rightarrow$ Protein complexes

$\Rightarrow$ **Lipoproteins**

Lipoproteins: classification based on density

- Chylomicrons
- Very low-density lipoproteins
- Intermediate-density lipoproteins
- Low-density lipoproteins
- High-density lipoproteins

Why do we lower Lipids?

➤ To lower risk of

1. Coronary heart disease

- ✓ ↑ed LDL associated with atherogenesis
- ✓ ↑ed HDL associated with ↓ed coronary vascular diseases

2. Pancreatitis

- ✓ Associated with ↑ed levels

- **Dyslipidemia** is elevated total cholesterol, low-density lipoprotein (LDL) cholesterol, or triglycerides; low high-density lipoprotein (HDL) cholesterol; or a combination of these abnormalities.

Classification of Lipid lowering Agents

- i. **HMG-CoA reductase inhibitors /statins/**
- ii. Fibrates: fibric acid derivatives
- iii. Bile acid Binding Resins (Sequestrants)
- iv. Nicotinic acid /Niacin/
- v. Ezetimibe

Inhibitors of cholesterol biosynthesis (Statins)

➤ MOA

➤ Potent competitive inhibitors of the enzyme 3-hydroxy-3-methylglutaryl coenzyme A reductase (HMG CoA reductase), the rate-limiting enzyme in cholesterol biosynthesis

➤ **Drugs:** Lovastatin, simvastatin, fluvastatin, pravastatin, atorvastatin, and rosuvastatin

- Drugs that inhibit cholesterol biosynthesis are quite effective at lowering LDL cholesterol and total cholesterol.

- This group of antihyperlipidemic agents inhibits the first committed enzymatic step of cholesterol synthesis, and they are the first-line and more effective treatment for patients with elevated LDL cholesterol.

Characteristic	<i>Atorvastatin</i>	<i>Fluvastatin</i>	<i>Lovastatin</i>	<i>Pravastatin</i>	<i>Rosuvastatin</i>	<i>Simvastatin</i>
Serum LDL cholesterol reduction produced (%)	50	24	34	34	50	41
Serum triacylglycerol reduction produced (%)	29	10	16	24	18	18
Serum HDL cholesterol increase produced (%)	6	8	9	12	8	12
Plasma half-life (hr)	14	1–2	2	1–2	19	1–2
Penetration of central nervous system	No	No	Yes	No	No	Yes
Renal excretion of absorbed dose (%)	2	<6	10	20	10	13

Statins (cont'd)

- Reduce free cholesterol content in hepatocytes.
- ↑ed expression of LDL receptors in hepatocytes
- Reduced degradation of LDL receptors
- ↑ed removal of LDL $\Rightarrow$ ↓ed LDL-C levels.
- **Modest decreases in plasma triglycerides and**
- **Small increases in HDL also occur.**
- **These drugs reduce total cholesterol by as much as 30%–50%; LDL cholesterol can be reduced by as much as 60% (rosuvastatin).**

Pharmacokinetics

- All given orally
- **Produgs:** lovastatin, simvastatin
- **Active drugs:** fluvastatin, atorvastatin, and rosuvastatin
- **All are extensively metabolized by the liver (first pass effect)**
 - low systemic bioavailability.
- **Most are highly plasma protein bound: > 95%**
- Elimination: mainly liver : feces.

➤ Adverse effects

- Usually well tolerated
- Mild GI complaints, muscle pain, rash, headache
- Myopathy
- Hepatotoxicity

➤ Contraindicated in pregnant & lactating women

➤ Use: effective in lowering plasma cholesterol levels in all types of hyperlipidemias.

- However, patients who are homozygous for familial hypercholesterolemia lack LDL receptors and, therefore, benefit much less from treatment with these drugs.

